

Foundations of Classical Mechanics for Molecular Dynamics (MD)

PWF Summer school on Theoretical and Computational Biophysics

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Day 1/5

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Why classical mechanics for biomolecules?

Biomolecules are large, dynamic many-body systems

- Proteins, water, lipids, ions and nucleic acids
- Typically, 10^4 – 10^7 atoms in realistic models
- We want motion, fluctuations and interactions
- Connect microscopic motion to measurable properties

atoms

interactions

motion

properties

Why classical mechanics for biomolecules?

- The motion of one atom depends on its interactions with many other atoms.
- At the fundamental level, a molecule consists of nuclei and electrons which are quantum mechanical particles.
- Electrons are much lighter than atomic nuclei and respond much more rapidly to nuclear motion.
- The **Born–Oppenheimer approximation** exploits this difference in timescales.
- Classical MD does not mean that the molecule itself is fundamentally classical. It means that we approximate atomic nuclei as classical particles moving on an effective potential energy surface.

Nuclear motion can be approximated classically

Electrons are treated implicitly through a potential-energy surface

- Classical MD generally goes one step further by using an empirical force field to approximate this energy surface.

Atoms as interacting classical particles

In classical molecular dynamics, we represent a system as:

$$\{m_i, r_i, v_i\}$$

- m_i — mass of atom i
- r_i — position vector
- v_i — velocity vector
- $U(r_1, \dots, r^N)$ — potential energy

MD computes $r_i(t)$ and $v_i(t)$ by repeatedly solving the equations of motion.

For N atoms, the molecular state is represented by $3N$ coordinates and $3N$ momenta

Classical MD does not track electronic wavefunctions during the trajectory.

Instead, it follows nuclei on an effective potential-energy surface.

Newton's laws: the mechanical foundation

First law — motion persists unless a net force acts.

$$F = ma$$

$$F_i = m_i \frac{d^2 r_i}{dt^2}$$

Third law — pair forces obey $F_{ij} = -F_{ji}$.

Position

$$r(t)$$

Velocity

$$v = dr/dt$$

Acceleration

$$a = d^2 r/dt^2$$

For MD, Newton's second law is the central equation.

From potential energy to force

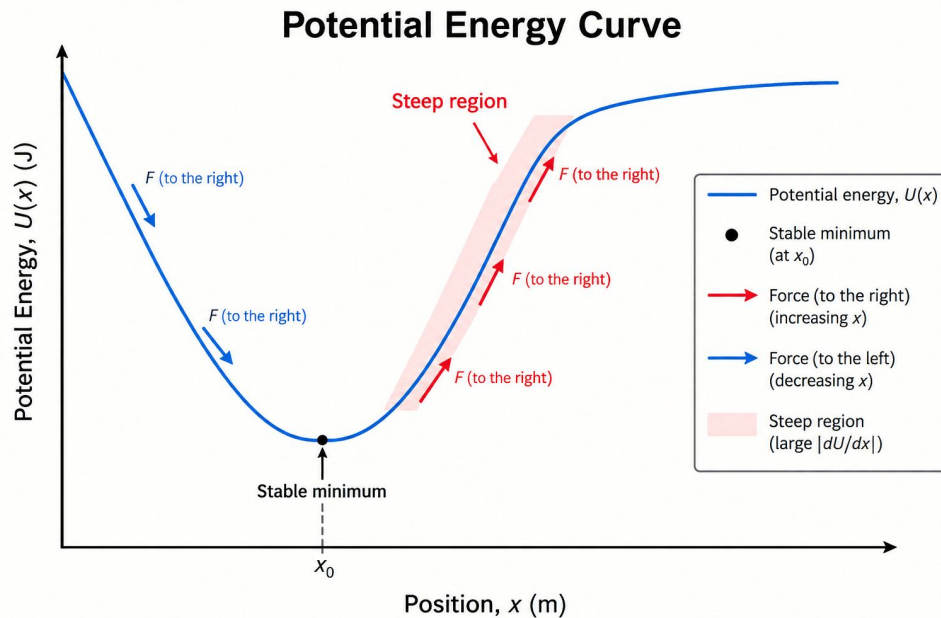
Potential energy
 $U(r)$

Gradient ∇U

Force F

$$F_i = -\nabla_i U$$

The force points downhill on the potential-energy landscape.



Example: Simulating a vibrating molecule

- At finite temperature, atoms have kinetic energy
- If the atom moves to either side of the stable minimum, a force directed back toward the minimum is produced.
- According to uncertainty principal, an atom cannot be at zero position and momentum simultaneously.
- Even at $T = 0$, an atom has **zero-point vibrational energy**. The uncertainty principle prevents simultaneously having an exactly defined position and exactly zero momentum.

$$U(x) \approx U(x_0) + \frac{1}{2}k(x - x_0)^2$$

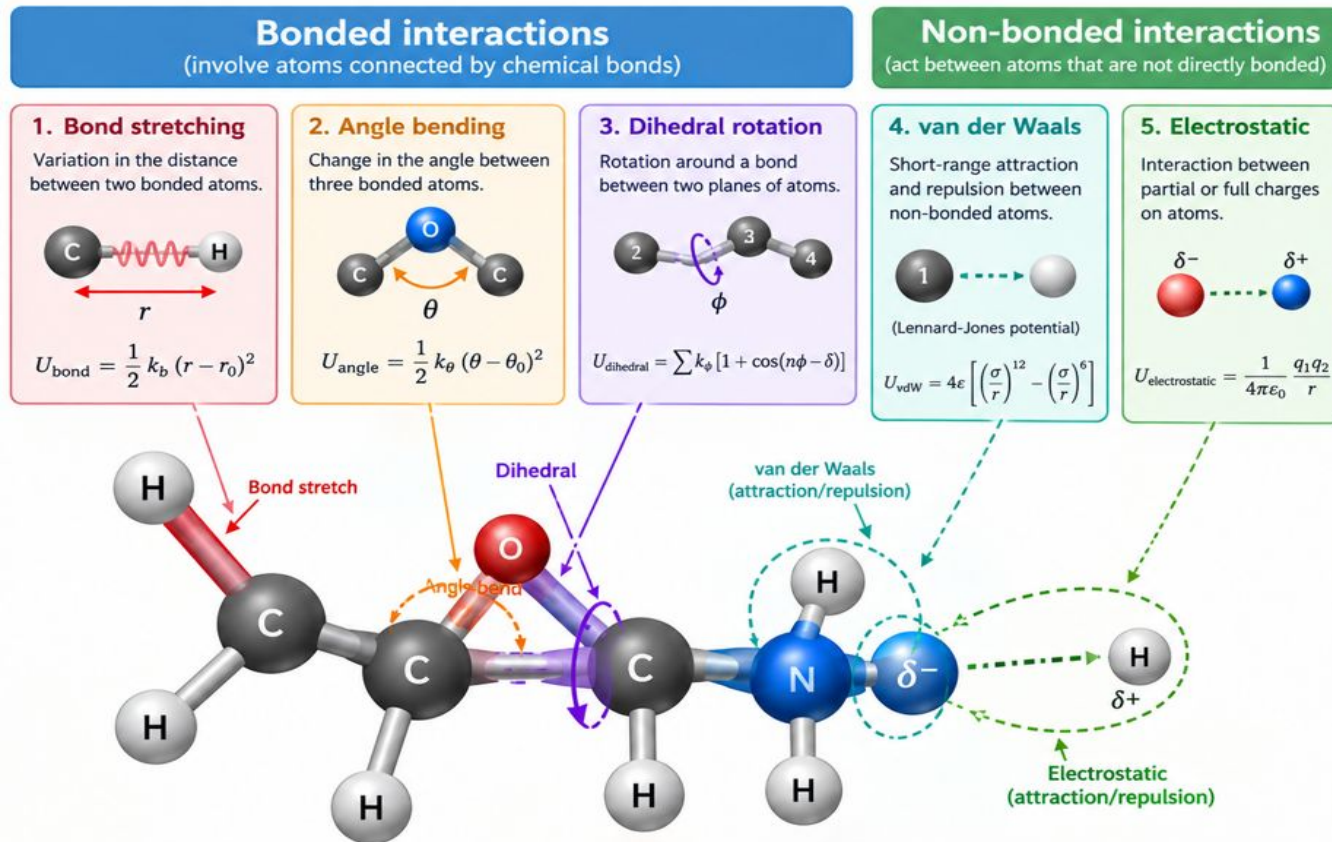
$$F = -k(x - x_0)$$



Bond vibrations can often be approximated locally by harmonic potentials

atoms are continuously moving around their equilibrium configurations rather than being fixed at the minima

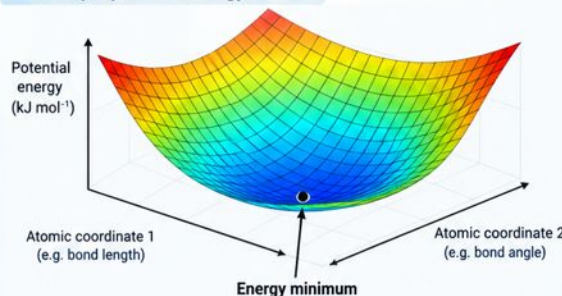
The molecular force field



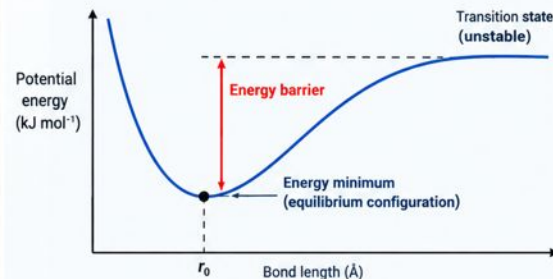
Potential Energy Surfaces (PES)

The potential energy surface describes how the potential energy of a system changes with the configuration of its atoms. Force-field parameters define this surface (e.g., bond terms, angle terms, dihedral terms, non-bonded terms).

1. A simple potential energy surface



2. 1D potential energy profile (e.g. bond stretching)



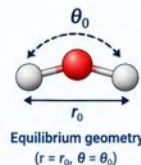
3. Energy minimum

- The lowest point on the potential energy surface.
- At this configuration the forces on all atoms are zero (stable equilibrium).



4. Equilibrium configuration

- The configuration corresponding to the energy minimum.
- For a bond, this is the equilibrium bond length (r_0) and angle (θ_0).



5. Energy barriers

- Peaks on the potential energy surface.
- Atoms must overcome these barriers to move from one configuration to another (e.g. rotation around a dihedral, chemical reactions).



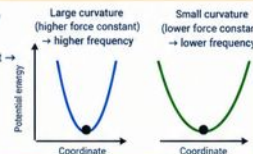
6. Stable vs unstable configurations

- Stable:** local minimum (positive curvature, forces restore the system).
- Unstable:** local maximum or saddle point (negative curvature, small displacement leads to motion away).



7. Curvature of the potential

- The curvature (second derivative) of the potential energy determines the force constant (stiffness).
- A steeper (larger curvature) well $\rightarrow$ larger force constant $\rightarrow$ higher frequency of vibration.



8. Why atoms vibrate rather than remain stationary?

- Even at the minimum, atoms have kinetic energy due to thermal motion.
- The potential energy near the minimum is approximately quadratic:

$$U \approx U_0 + \frac{1}{2} k(x - x_0)^2$$

- This leads to harmonic motion (vibrations) around the equilibrium position.
- At finite temperature, atoms vibrate with an amplitude determined by the temperature (and mass and force constant).



In summary: Force-field parameters shape the potential energy surface, which determines where atoms prefer to be, how they move, and how they interact.

Force Field Parameters

A force-field parameter is a numerical value that determines the strength, preferred geometry, or functional form of an interaction in the potential-energy model.

Where do force-field parameters come from?

Experimental data

- bond lengths
- vibrational frequencies
- densities
- heats of vaporization
- dielectric properties
- structural data

Quantum-mechanical calculations

- equilibrium geometries
- energy surfaces
- electrostatic potentials
- torsional energy profiles
- vibrational properties

Optimization/fitting

Parameters are adjusted to reproduce selected experimental or QM properties.

The equations we have just discussed eventually become text files.

Example of a parameter file

Text of the slide

The molecular dynamics algorithm

1 Positions
 $r(t)$

2 Calculate
 $U(r)$

3 Calculate $F = -\nabla U$

4 Calculate $a = F/m$

5 Update r and v

6 Repeat

One simulation = millions of repeated force evaluations and integration steps.

Why numerical integration?

Many-body molecular equations rarely have analytic solutions.

We approximate the trajectory over small time intervals.

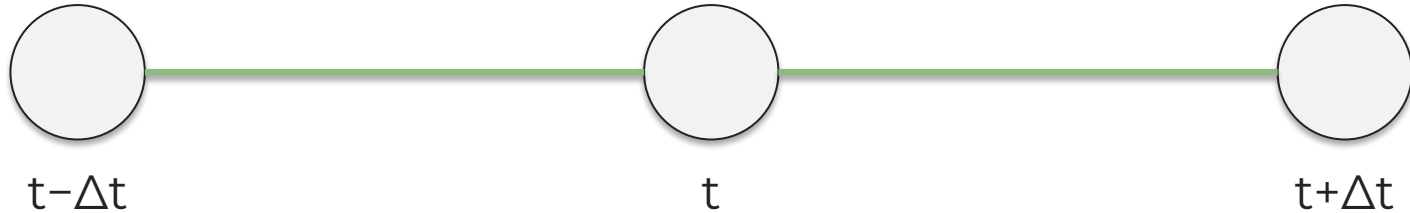
$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t + \frac{1}{2}\mathbf{a}(t)\Delta t^2 + \dots$$

Common integrators

- Verlet
- Velocity Verlet
- Leapfrog

Verlet integration

$$r(t+\Delta t) = 2r(t) - r(t-\Delta t) + a(t)\Delta t^2$$



Uses positions at neighbouring time steps; simple and time-reversible.

Velocity Verlet

$$r(t+\Delta t) = r(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2$$

$$v(t+\Delta t) = v(t) + \frac{1}{2}[a(t)+a(t+\Delta t)]\Delta t$$

1. Update position

2. Recalculate forces

3. Update velocity

Positions and velocities remain synchronized with the force calculation.

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